# CLEARING THE PEDIATRIC C-SPINE

## 1 FLOWCHARTS

CHILD WITH SUSPECTED C-SPINE INJURY

History
- Persistent neck pain
- Abnormal head posture
- Difficult head movements
- Focal sensory abnormality or motor deficit

Physical Examination
- NEXUS criteria
  - Neurological deficit
  - Posterior midline neck tenderness
  - Altered Mental State
  - Intoxication
  - Distracting injury
- Torticollis / abnormal head position
- Limited cervical ROM

**Stronger consideration for imaging should be given to patients with pre-disposing risk factors or high risk mechanism of injury should

"NO" TO ALL OF THE ABOVE → C-SPINE CLEAR!

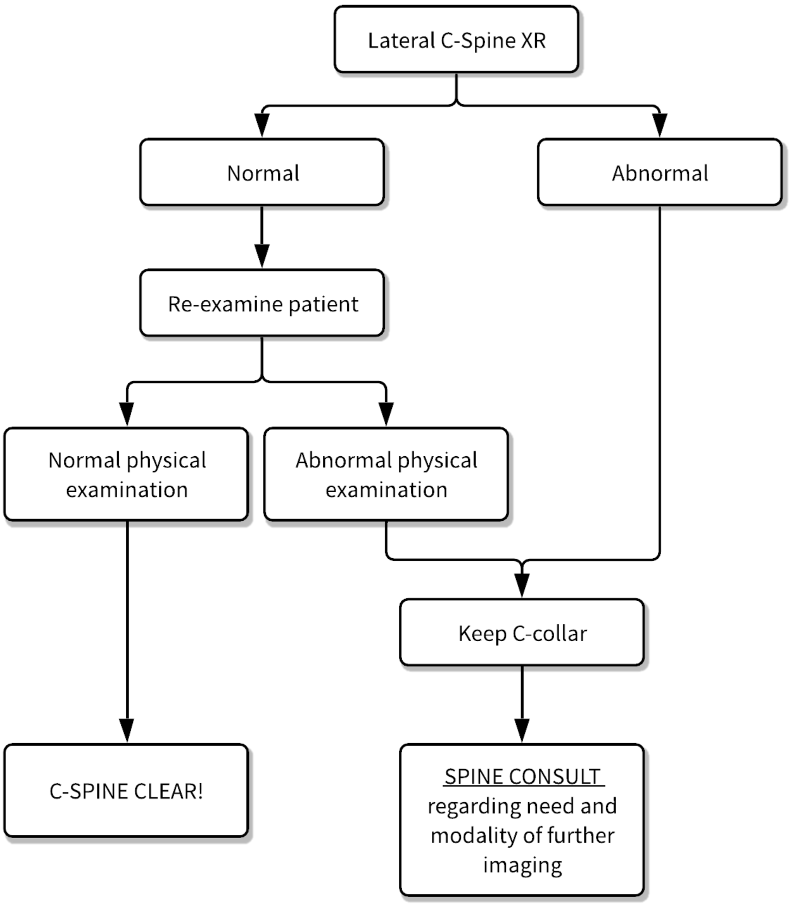
"YES" TO ANY OF THE ABOVE → Lateral C-Spine XR#

High Risk Mechanisms
1. Axial load
2. Fall > 3 metres or 2x height
3. Diving
4. High risk motor accident
   - Speed > 60km/h
   - Death in the same vehicle
   - Ejected from vehicle
   - Head on collision
   - Rollover

Pre-disposing risk factors
1. Trisomy 21
2. Achondroplasia
3. Previous c-spine injury

#Choice of radiologic imaging
- Lateral c-spine view
  - Initial radiographic evaluation for ALL children clinical clearance is not possible
  - Other views (AP, odontoid, oblique, flexion / extension) are unnecessary for initial evaluation
- CT c-spine
  - Patients with AMS
  - Patients planned for CT Brain

*continued on page 2*




## 1.1 CERVICAL SPINE MOTION RESTRICTION

- Suspect cervical spine injury in any patient with multi-system trauma.
- Ability to walk away from the scene <u>does not</u> negate the need for cervical collar.
- Cervical spine motion restriction <u>with a rigid cervical collar</u> should be maintained until cervical spine injury has been excluded clinically and/or radiographically.
- Contraindications to a rigid cervical collar:
    - Fixed neck deformity.
    - Massive cervical swelling for which placing a collar could further compromise airway.
    - Need for emergent surgical airway (e.g., cricothyrotomy)
    - Worsening neurology after placement of collar.
- Refer to **<u>ANNEX A</u>** for sizing and **<u>ANNEX B</u>** for application of pediatric cervical collars.

## 1.2 NEUTRAL POSITION OF THE SPINE

- The neutral position of the spine maximizes the spinal canal diameter → reduces further injury to the spinal cord.
- When the cervical spine is in neutral position (this applies to morphologically normal patients only)
    - Patient's gaze would be perpendicular to the surface.
    - External auditory meatus is aligned with the shoulder in the coronal plane.
- Patients > 8-years require occipital elevation to maintain the neutral position.
- Younger patients and infants have relatively larger heads and require some elevation from shoulders inferiorly to achieve a neutral position of the spine.

# 2 PATTERN OF INJURY & ANATOMICAL CONSIDERATIONS

## 2.1 PATTERN OF INJURY

- Age < 8-years
  - Injuries are typically of the upper cervical spine (C2 and above).
  - Young spinal column is more elastic than the spinal cord → able to tolerate more distraction before rupture → can have spinal cord injury without radiographic evidence of spinal column injury.
- Age ≥ 8-years: Lower cervical spine vertebral body and arch fractures

## 2.2 ANATOMICAL FEATURES (CHILDREN < 8-YEARS)

- Relatively larger and heavier heads.
- Anatomical "fulcrum" progresses caudally from C2/C3 (at birth) to C5/C6 (by 8-years-old)
- Ligaments and joint capsules are more lax → greater mobility of the upper cervical spine.
- Immature vertebral joints & horizontally inclined articulating facets that facilitate sliding of the upper cervical spine.

## 2.3 TYPES OF INJURY

| FRACTURES | SUBLUXATION & DISLOCATION |
|---|---|
| **Atlas # (C1)** a.k.a. Jefferson #<br>▫ 2 patterns of injury: Posterior arch # (more common) & burst #<br>▫ Result of axial loading.<br>▫ Neck pain + Limited ROM<br>▫ Usually normal neurological examination<br>▫ Diagnosed on open-mouth odontoid views | **Cervical vertebral subluxation**<br>▫ Axial compression + cervical flexion (sports injury)<br>▫ Posterior ligament injury<br>▫ Anterior translation of the superior vertebral body |
| **Axis # (C2)**<br>Odontoid #s<br>▫ Direct head impact<br>▫ # at the tip or the base<br><br>Hangman's # (spondylolisthesis of C2)<br>▫ # of pedicle of C2<br>▫ Hyper-extension injury during RTA<br>▫ Usually normal neurological examination | **Acute atlanto-axial instability (AAI)**<br>▫ Extreme flexion of the neck<br>▫ Injury to transverse ligament of the odontoid process<br>▫ Posterior translation of odontoid compress spinal cord<br>▫ Diagnosed on lateral neck XR with widened atlanto-dental interval (ADI) [*see section 3.2*] |
| **Compression #**<br>▫ Result from axial loading (± flexion or extension)<br><br>**Cervical spinous process #**<br>▫ Avulsion # of the tip of spinous process<br>▫ C7 is most commonly affected<br>▫ Vigorous labour with forceful contraction of shoulder muscles | **Atlanto-axial rotary subluxation (AARS)**<br>▫ Rotational displacement of C1 on C2<br>▫ Retropharyngeal edema → laxity of ligaments at atlanto-axial level → rotary deformity<br>▫ Age group: 6 – 12 years<br>▫ Presents with pain / torticollis (head tilts away from affected SCM)<br>▫ Tenderness over spinous process of axis with displacement towards side of torticollis<br>▫ C2 nerve runs in capsule of atlantoaxial joint → unilateral occipital pain |

# 3 Radiology

## 3.1 Reading the C-Spine Radiograph

| | |
|---|---|
| Coverage | All c-spine vertebrae present including C7/T1 junction |
| Alignment | Lines must be smooth with no steps or angulation. [*Figure 1*]<br>1. Anterior vertebral line (**red**)<br>2. Posterior vertebral line (**green**)<br>3. Facet line (**blue**)<br>4. Spino-laminar line (**yellow**) |
| Bones | Check for breaks in cortical outline of all vertebrae/bones. [*Figure 2*]<br>C1 (atlas) – Anterior arch of atlas<br>C2 (axis) – Odontoid peg of C2<br><br>Developmental variant<br>▫ Anterior narrowing of the vertebra can mimic compression #.<br>▫ Slightly wedged shaped vertebral body common in adolescence (C3)<br><br>Intervertebral disc spaces should be equal. |
| Pre-vertebral space | The pre-vertebral space is considered abnormal if: [*Figure 3*]<br>▫ C2 > 7mm<br>▫ C6 > 14mm (for adults: > 22mm)<br>This could be secondary to:<br>▫ Vertebral fracture with resultant adjacent hematoma (in trauma)<br>▫ Edema from a deep neck infection (non-trauma) e.g., retropharyngeal abscess<br><br>*Generally, should be one-half the width of the adjacent vertebral body |

## 3.2 Important Measurements

| | |
|---|---|
| Atlanto-Dental Interval (ADI) | ADI > 5 mm (> 2.5 mm, adults) is abnormal and indicates atlanto-axial instability (AAI). [*Figure 3*] |
| Pseudo-subluxation | C2 on C3 (most common)<br>▫ Spino-laminar line remains smooth even in the presence of pseudo-subluxation |

*Figure 2*

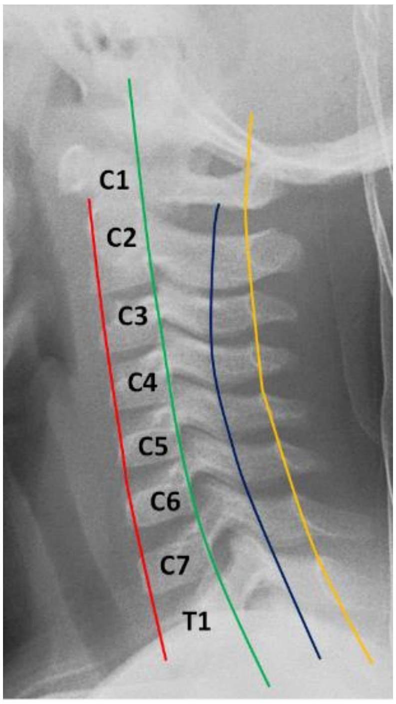



*Figure 1*

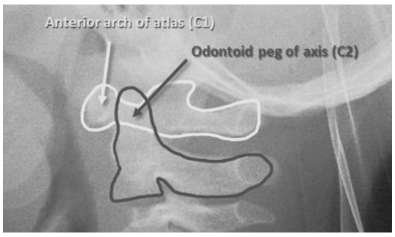
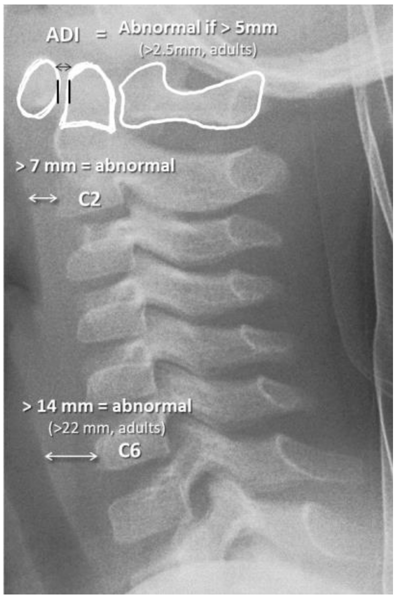


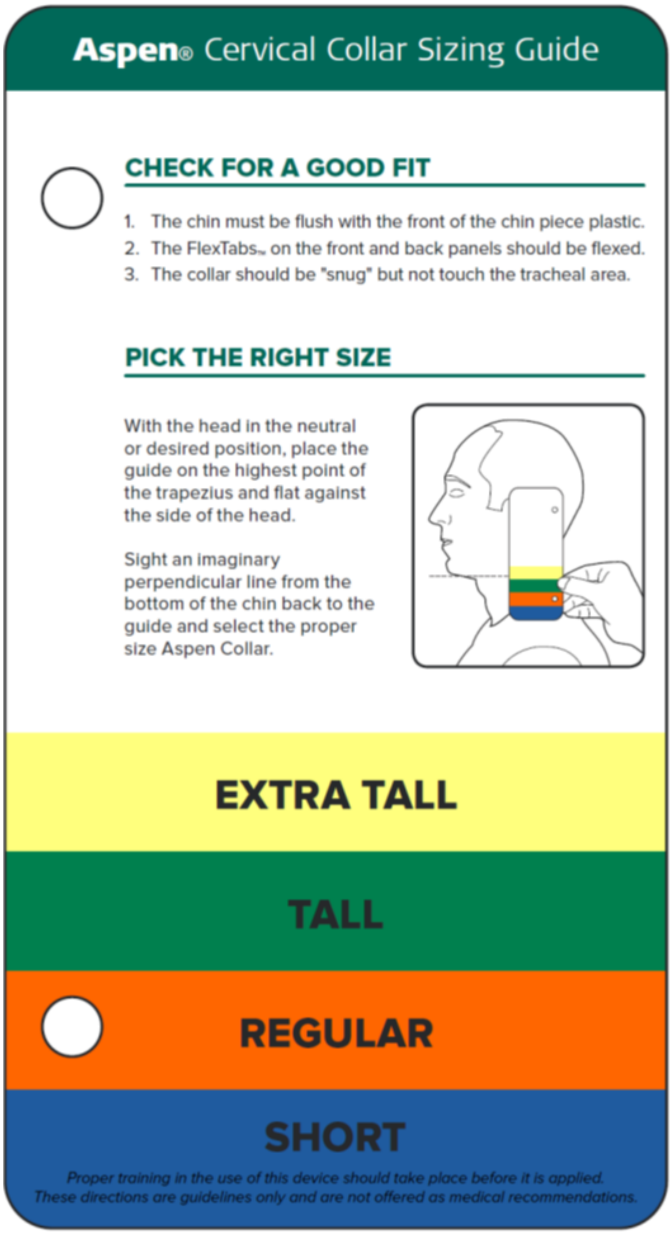

*Figure 3*

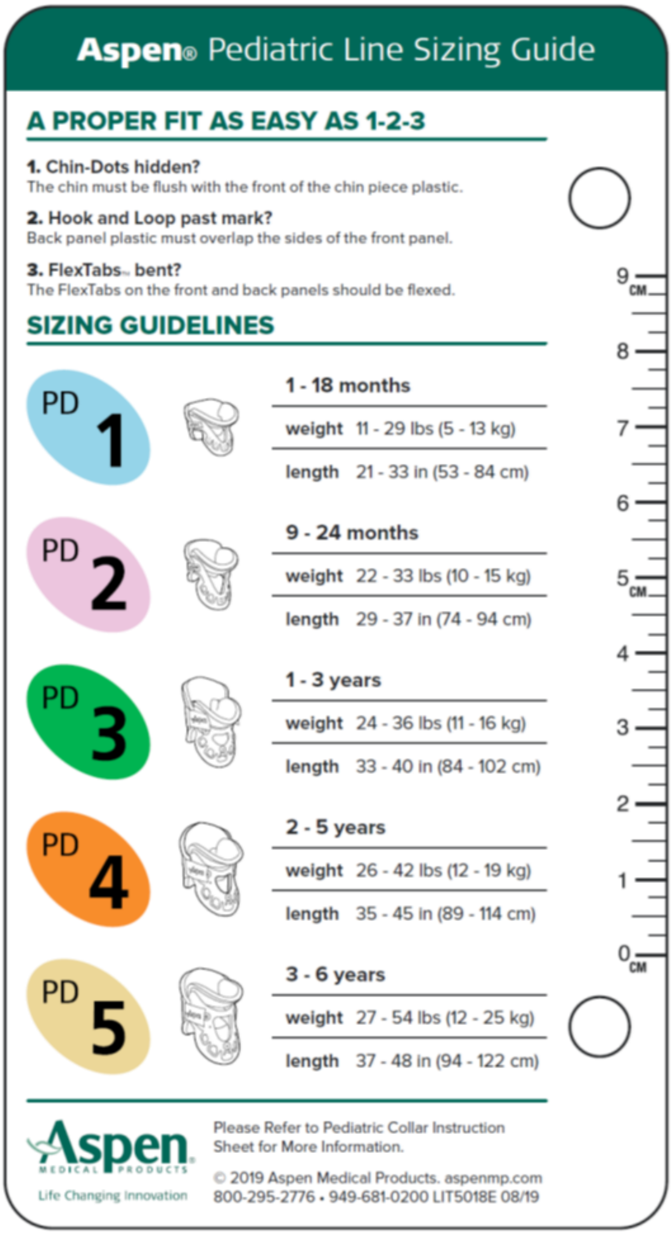


# ANNEX A. SIZING THE CERVICAL COLLAR

- Too tall a collar can cause hyperextension.
- Too short a collar can affect the airway and compromise on motion restriction.

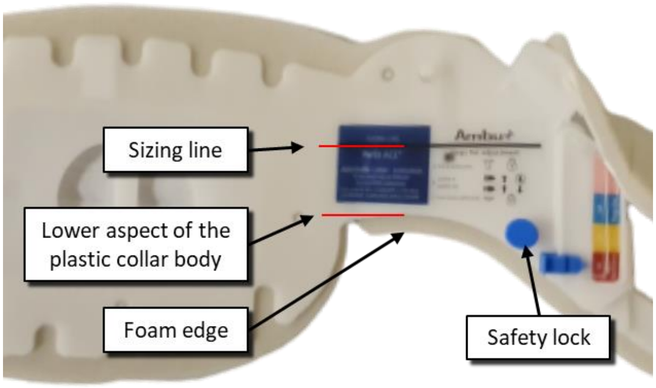
## I. ASPEN® PEDIATRIC CERVICAL COLLAR



## II. AMBU® PERFIT® ACE®

- Measure the distance between an imaginary horizontal plane below the patient's chin to the highest part of the trapezius muscle.
- Compare the distance with the distance from the collar sizing line to the lower aspect of the plastic collar body (not the foam)
- Disengage the safety locks by pulling UP, adjust the collar, engage the safety locks by pushing DOWN.



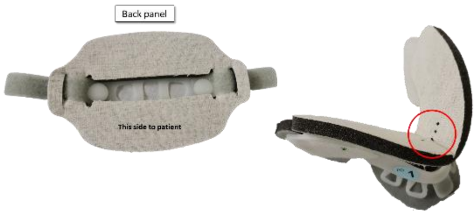
# ANNEX B. APPLICATION OF CERVICAL COLLAR

## I. Applying the Aspen® Pediatric Cervical Collar

- Age range supported: 1-month to 6-years of age.
- Five different colour-coded sizes, PD1 to PD5.
- Three paediatric sizes (PD3 to PD5) and two infant sizes (PD1 to PD2).

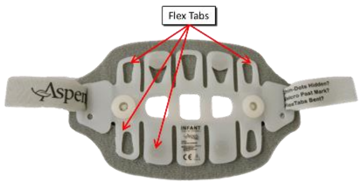
<u>APPLYING THE ASPEN® PEDIATRIC CERVICAL COLLAR</u>

1. Positioning
   a. Ensure patient in neutral spine position.
   b. Provide in-line stabilization if necessary.
   c. Slide back panel behind patient's neck.
   d. Squeeze the front panel in half horizontally and scoop the panel up and under the chin, aligning patient's chin with anterior border of the plastic chin piece
   e. Chin-Dots (red circle) should be covered




2. Securing
   a. Hold trach hole with one hand.
   b. Secure back panel flap onto the side of the front panel.
   c. Repeat on opposite side and alternate sides till collar is secure and centred.
   d. Flaps should extend past the green mark (red circle)

3. Checking the fit
   a. If collar has been properly sized and tightened, Flex Tabs should be bent/flexed




## II. Ambu® Perfit® ACE®

- Proper application requires 2 individuals, one for in-line stabilization and another for application of cervical collar.

<u>APPLYING THE AMBU® PERFIT® ACE®</u>

1. Slide the back of the collar behind the patient's neck.
2. Position the chin piece.
3. Secure with Velcro strap, ensuring that the loop Velcro tab is firmly attached and parallel to the fixed hook Velcro tab on the front of the collar.

# 4 REFERENCES


1. "A Consensus Statement and Algorithm from the Pediatric Cervical Spine Clearance Working Group" Pediatric Cervical Spine Clearance, The Journal of Bone and Joint Surgery: January 2, 2019 - Volume 101 - Issue 1 - p e1. doi: 10.2106/JBJS.18.00217 Herman, Martin J. et. al.
2. "Clearing the cervical spine of paediatric trauma patients" Emergency Medicine Journal 2004;21:189-193. Slack SE, Clancy MJ et al.